NAME: ____________________ **PERIOD:** ________ **DATE:** ____________

Two Samples of Service Times

AP Statistics · Unit 4 · Topic 4.7 · B: 2027-02-08 · E: 2027-02-10

[STAR] TODAY'S PROBLEM

Suppose two depots independently take SRSs of 36 weekday service-time records from monthly populations of 900 North and 1,200 South records. Technology gives $df = 69.31$ and $t^* = 1.995$ for a 95% two-sample interval.

Tariq: "I sorted both samples and paired equal ranks. The resulting differences have $\bar{d} = 2.2$ and $s_d = 3.0$, giving $(1.18, 3.22)$. Pairing makes the estimate more precise."

Mei: "The records were sampled separately, so I would use a two-sample interval. Sorting observed responses may change the uncertainty rather than reveal design-based pairs."

Service-time samples (min)

Depot	N	n	$\bar{x}$	s
North	900	36	42.3	8.4
South	1,200	36	40.1	7.6

FIRST TAKE — before we discuss (5 min)

Does sorting the times make the North and South records real pairs? Explain in one sentence.

[BOOK] LET THE DESIGN CHOOSE THE INTERVAL (keep on the page)

For two independent random samples, use a two-sample t interval: $(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{s_1^2/n_1 + s_2^2/n_2}$. Paired measurements have a link in the study design, such as two measurements from the same person; they use one sample of differences. Check independent random samples or assignment, each 10 percent condition for sampling without replacement, and both sample shapes or both sample sizes over 30. Interpret in the stated subtraction order. An interval containing 0 leaves equal population means plausible; an interval entirely on one side supports a difference.

Work (a)–(c) from the rules box:

DOK 1 **(a)** Find the North-minus-South difference in sample mean service times.

DOK 2 **(b)** The independent-sample standard error is 1.88797 minutes. Use $t^* = 1.995$ to calculate the 95% interval.

DOK 3 **(c)** In 3–4 sentences, choose a procedure using the original sampling design. Compare the two intervals and explain how sorting the times could mislead a reader about whether the population means differ.

Frame: The design supports a ____________________ interval because ____________________________; sorting ranks instead _________________________.

Frame: The two uncertainty estimates are ____________________ and ____________________, which changes _________________________.

EXIT — turn this in

One thing the rules box changed about my first take: ____